

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer, WordNetLemmatizer

from nltk import pos_tag, FreqDist


# Download required resources (run once)

nltk.download('punkt_tab')

nltk.download('stopwords')

nltk.download('wordnet')

nltk.download('averaged_perceptron_tagger')

nltk.download('omw-1.4')

nltk.download('averaged_perceptron_tagger_eng')

# Input resume text

text = input("Enter resume text: ")


# 1. Tokenization

tokens = word_tokenize(text)


# Keep only alphabetic words and convert to lowercase

words = [word.lower() for word in tokens if word.isalpha()]


print("Tokens:")

print(words)


# 2. Stopword Removal

stop_words = set(stopwords.words('english'))

filtered_words = [word for word in words if word not in stop_words]
```

```
print("\nAfter Stopword Removal:")
```

```
print(filtered_words)
```

3. Stemming

```
stemmer = PorterStemmer()
```

```
stemmed_words = [stemmer.stem(word) for word in filtered_words]
```

```
print("\nStemmed Words:")
```

```
print(stemmed_words)
```

4. Lemmatization

```
lemmatizer = WordNetLemmatizer()
```

```
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]
```

```
print("\nLemmatized Words:")
```

```
print(lemmatized_words)
```

5. POS Tagging

```
pos_tags = pos_tag(lemmatized_words)
```

```
print("\nPOS Tags:")
```

```
print(pos_tags)
```

6. Frequency Distribution

```
freq_dist = FreqDist(lemmatized_words)
```

```
print("\nFrequency Distribution:")
```

```
for word, freq in freq_dist.items():  
    print(word, ":", freq)
```